

Transcriptomic Analysis of Organ-Specific Metastatic Adaptation in Breast Cancer

Background:

Metastasis is a context-dependent process influenced by both tumor-intrinsic properties and the target organ microenvironment. Understanding how cancer cells adapt transcriptionally to different metastatic niches remains a key challenge.

Methods:

Public microarray data (GSE62598) from organ-tropic metastatic breast cancer sublines (4T1 model) were analyzed. Probe-level data were mapped to gene symbols and collapsed to gene-level expression. PCA was performed on the top 1000 most variable genes. Differential expression analysis and pathway enrichment (KEGG) were conducted.

Key Findings:

- PCA revealed partial separation of liver-tropic cells along a dominant transcriptional axis.
- Liver and lung metastatic sublines showed relatively modest transcriptional differences, while liver vs bone showed greater divergence.
- No genes passed strict statistical thresholds due to small sample size.
- Exploratory analysis identified hundreds of differentially expressed genes under relaxed thresholds.
- Pathway enrichment consistently highlighted metabolic processes, including lipid metabolism, amino acid metabolism, and central carbon metabolism.

Interpretation:

These results suggest that metastatic adaptation is not driven by large gene-level changes but rather by coordinated pathway-level shifts. In particular, liver-tropic cells show evidence of metabolic reprogramming, consistent with adaptation to a metabolically distinct microenvironment. This supports a model in which organ-specific metastatic fitness is shaped by context-dependent biological constraints.

Conclusion:

This analysis highlights the importance of pathway-level interpretation in metastasis research and supports the hypothesis that metabolic adaptation plays a central role in organ-specific colonization.